Indian Institute of Information Technology Kottayam

IEG 313: Quantum Computing for Engineers Theory and Applications
Project Title and Summary Submission

Project Group Name: Dirac's Army
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Project Title

Quantum-Assisted Diagnosis: Projected Quantum Kernel-Enhanced CNNs for Rare Disease Detection in Data-Scarce Medical Imaging

Abstract / Project Summary

Spotting rare conditions on medical scans is notoriously difficult for AI. Standard deep learning models thrive on massive volumes of data, but with rare diseases, doctors might only have a handful of confirmed cases to work with. When fed such tiny datasets, standard convolutional neural networks (CNNs) tend to memorize the training images rather than learn meaningful patterns, causing them to fail in real clinical settings right when early detection matters most.

To tackle this shortage of data, this project explores a hybrid classical-quantum approach. Instead of relying purely on conventional methods, a standard CNN handles the initial heavy lifting by pulling out key clinical features from the images. Those features are then translated into a few qubits and evaluated through a Projected Quantum Kernel (PQK). By projecting the data into a high-dimensional quantum space, the model can catch subtle, non-linear relationships that classical algorithms overlook giving it an edge when working with sparse data.

The system will be built and simulated in PennyLane and TensorFlow, then benchmarked against a conventional classical CNN using a restricted sample of scans (such as a downsampled slice of ChestX-ray14 or a specialized rare-disease dataset). The goal is to show that a quantum-boosted model can noticeably outperform traditional baselines when labeled data is scarce, providing a practical proof-of-concept for quantum-assisted diagnostics where clinical data is hard to come by.

References
List the major references/resources relevant to the proposed work.

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[1] "Enhancing Small Dataset Classification Using Projected Quantum Kernels with Convolutional Neural Networks," arXiv:2601.03375, Jan 2026.

[2] PennyLane Quantum Machine Learning Demos — "Quantum Kernels" tutorial, PennyLane.ai documentation.

[3] Wang, X. et al., "ChestX-ray14: Hospital-scale Chest X-ray Database," NIH Clinical Center dataset.

Instructions:

1. Each team should submit only one document.
2. Clearly mention the project title and the names/roll numbers of all team members.
3. The abstract/project summary should briefly explain what the project is about, what you plan to do, and the expected outcome.
4. Keep the abstract concise, preferably 150–250 words.
5. Avoid copying text directly from online sources or research papers.
6. Use a clear and specific project title.
7. Submit the document in PDF (.pdf) format.
8. File name: TeamName.pdf
9. Submission deadline: 23-09-26 , 11:59pm